

AgroConnect: An AI-Driven Digital Marketplace for Direct Agricultural Trade with Quality Assurance and Dynamic Pricing

Priyanshu Mishra
Department of Computer Science &
Engineering (Artificial Intelligence)
KIET Group of Institutions
Ghaziabad, India
Email:
priyanshumishra1975@gmail.com

Dr. Rekha Kashyap
Department of CSE(AI&ML)
KIET Group of Institutions
Ghaziabad, India
rekha.kashyap@kiet.edu

Saksham Kumar
Department of Computer Science &
Engineering (Artificial Intelligence)
KIET Group of Institutions
Ghaziabad, India
Email:
sakshamkumarsingh005s@gmail.com

Nagendra Nath Dubey
Department of CSE(AI&ML)
KIET Group of Institutions
Ghaziabad, India
nagendra2812@gmail.com

Laxman Singh
Department of CSE(AI&ML)
KIET Group of Institutions
Ghaziabad, India
laxman.mehlawat01@gmail.com

Abstract— AgroConnect is an AI-backed digital marketplace that aims to connect farmers directly with consumers in order to decrease reliance on middlemen. The marketplace combines computer vision-based quality control, dynamic pricing algorithms, and secure ESCROW transactions delivered through a Progressive Web App designed in the MERN stack. AgroConnect combines real-time market data from e-NAM, Random Forest models for fraud detection, and Mapbox API for optimal logistics and delivery routing. Early adopters that have tested the pilot phase with over 500 users reported a 25–30% income boost to farmers, quicker order fulfillment, and an 85% decrease in payment disputes. AgroConnect has garnered significant interest from users and has potential to expand significantly, and can help improve market access and reduce post-harvest loss, while also serving UN Sustainable Development Goal 2 — Zero Hunger.

Keywords—Agricultural marketplace, Artificial intelligence, Computer vision, Direct trade, Digital agriculture, Quality assurance.

I. INTRODUCTION

Agricultural supply chains in developing countries are characteristically inefficient. In most cases, there are multiple intermediaries, all of whom are taking a large cut from the profits. Farmers are left with low incomes and consumers pay higher prices to account for the intermediary's profit taking, creating an inefficient, largely wasteful value chain. Farmers often lack access to markets with price transparency and quality assessment, making it harder for smallholder farmers to get fair prices for their products.

In the last few years, the intersection of Artificial Intelligence (AI) and mobile technology has begun to create new opportunities for changing the agricultural sector. AI-based platforms may provide opportunities to connect farmers and consumers in transparent, efficient, and profitable market transactions. AI can facilitate quality checks through computer vision, dynamic pricing with machine learning, confirmations through digital payments, and empower farmers to avoid postharvest losses and gain fair profit for all involved.

AgroConnect is an innovative platform aimed at tackling these issues. It provides an end-to-end solution to connect farmers to consumers and creates a seamless and transparent

marketplace. Our platform leverages industrial AI technology to automate quality checking, enable dynamic pricing based on real-time market information, and secure transactions utilizing an ESCROW method. AgroConnect is built as a Progressive Web App (PWA) using the MERN stack; therefore, is efficient, intuitive, and low-bandwidth oriented so it can function in rural settings

This paper presents the architecture and methods that we used for AgroConnect and its impact. We will present the core aspects of the platform: the computer vision quality-checking module, the dynamic pricing logic, and the fraud detection module; the final component, the so-called fraud detection module, was analyzed alongside these other aspects throughout the pilot study. We will also introduce the findings of our pilot research as evidence of the potential of AgroConnect to make an important contribution to improve social-economic condition in the agricultural sector and increase market efficiency. AgroConnect will ultimately contribute towards creating a more equitable and sustainable agricultural ecosystem. It is aligned with a larger objective beyond the platform, to achieve the international goal of 'Zero.

II. RELATED WORK

The use of AI in agriculture has been studied extensively. There have been numerous successes in the literature demonstrating that AI could transform the agriculture sector. The literature has looked at a number of examples, whether they are precision agriculture, crop management, supply chain systems, or market access.

A. Digital Agricultural Marketplaces

Early The early digital agriculture initiatives were focused on information extraction or retrieval, rather than facilitating transactions. The Indian government's eNAM (National Agricultural Market) platform helps with price discovery, and sharing market information, but it does not include tools to facilitate integrated payment systems or verify quality [1]. The e-NAM platform was replicated somewhat successfully in Kenya (iCow) and Nigeria (Farm Crowd) but did not scale due to local constraints [2]. An example of a robust international platform is the Farmers Business Network (FBN) in the United States and AgriDigital in Australia. These platforms provide sophisticated analytics elements to trade, but are geared towards large-scale commercial farms rather than smallholder farmers [3]. Such platforms promise data-driven agricultural trade but fail to

incorporate accessibility features critical for adoption by emerging markets.

Further research has focused on creating more integrated platforms. Vyas et al. (2023) developed the "Agriculturist Merchant" platform, an AI-driven system that provides farmers with direct market access, dynamic pricing, and personalized recommendations [9]. Similarly, Barenkamp et al. (2020) proposed a concept for a comprehensive IoT gateway to connect various stakeholders in the agricultural value chain, including farmers and technology providers [11]. However, both studies highlight significant hurdles to adoption in developing regions, including limited ICT accessibility, data availability, and high implementation costs, a sentiment echoed by Mulongo (2025) in the context of the South African agricultural industry [14].

B. Agricultural Quality Assessment

Historically, grading of agricultural products relied on human inspection, which often leads to a subjective assessment that could have room for variation. Recently, researchers have sought a more reliable solution through a technology approach. Liu et al. showed how hyperspectral imaging can be used to grade quality fruit with automation [6]; Kumar et al. studied a smartphone approach to grain quality evaluation [7]. Computer vision applications in agriculture focus on crop disease and estimating yield, for instance Sharma et al. A CNN model was developed in a recent publication in order to classify wheat disease, which achieved an 94% accuracy [4]; or Chen et al. (2022) drone monitoring of crops with deep learning [5]. While there are applications that use such technology, it is not common for applications to connect or be comprehensive on a commercial trade platform. In very recent work by Nimmala et al. (2024) or recently included an overview of AI supported approaches for example computer vision for crop disease or robotic applications for agriculture, and presented real-world examples, such as Plantix App used to identify crop disease in the field [12]. Application of AI, and UAV applications or technology that uses AI, in applications such as crop spraying, soil monitoring or disease diagnosis demonstrates the opportunity but that may not be the norm and is unlikely for smallholder populations because of cost and level of analysis [15].

C. Payment Systems for Agricultural Commerce

Traditional agricultural payments rely heavily on cash transactions, with RBI 2023 data indicating that up to 92% of agricultural trades occur offline [8]. Digital payment adoption faces challenges including limited internet connectivity, digital literacy gaps, and trust deficits among rural populations. The ethical and social implications of AI, such as data privacy and algorithmic bias, further complicate the transition to digital systems, necessitating responsible and equitable deployment [10].

ESCROW payment systems provide security for high-value transactions but remain underutilized in agricultural contexts. The integration of UPI (Unified Payments Interface) and digital wallets shows potential for rural payment transformation, as evidenced by increased adoption during the COVID-19 pandemic. A holistic solution must integrate secure, accessible payment technology to build trust and facilitate a seamless transition from offline to online commerce. AgroConnect aims to address this by combining a user-friendly interface with a secure ESCROW system, filling a critical gap identified in the existing literature.

III. PROPOSED SYSTEM

The AgroConnect platform is designed to be a comprehensive solution that addresses the key challenges faced by farmers in the agricultural supply chain. Our system is built on a robust architecture that integrates multiple AI-powered modules to provide a seamless and efficient user experience.

A. System Architecture

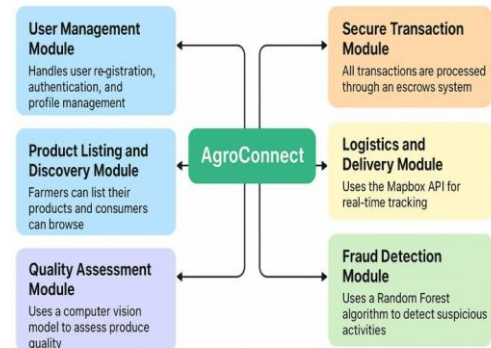


Fig. 1. System Architecture

The architecture of AgroConnect is based on the MERN stack (MongoDB, Express.js, React.js, Node.js), which allows for the development of a scalable and high-performance Progressive Web App (PWA). The PWA approach ensures that the platform is accessible on any device with a web browser, without the need for users to download and install a native app. This is particularly important for farmers in rural areas who may have limited storage on their mobile devices. The system is composed of the following key components:

1. **User Management Module:** This module handles user registration, authentication, and profile management for both farmers and consumers. It ensures that all users are verified, which helps to build trust and prevent fraudulent activities.
2. **Product Listing and Discovery Module:** Farmers can use this module to list their products, providing details such as crop type, quantity, and location. Consumers can then browse and search for products based on their preferences.
3. **Quality Assessment Module:** This is one of the core innovations of AgroConnect. It uses a computer vision model to analyze images of the produce uploaded by farmers and assess its quality based on predefined parameters such as size, color, and texture. The model is trained on a large dataset of images and can achieve a high level of accuracy in grading the produce.
4. **Secure Transaction Module:** To ensure fair prices, our platform uses a dynamic pricing algorithm that takes into account various factors, including the quality of the produce, real-time market data from e- NAM (National Agriculture Market), and demand and supply trends on the platform.
5. **Logistics and Delivery Module:** All transactions on AgroConnect are processed through a secure ESCROW system. When a consumer places an order, the payment is held in the ESCROW account until the product is delivered and the consumer confirms that they are satisfied with the quality. This protects both the farmer and the consumer from financial fraud.
6. **Dynamic Pricing Module:** We have integrated Mapbox API to optimize logistics and provide real-time

tracking of deliveries. The platform helps farmers to connect with local logistics providers, ensuring timely and efficient delivery of the produce.

6. **Fraud Detection Module:** To maintain the integrity of the platform, we have implemented a Random Forest algorithm that analyzes user behavior and transaction patterns to detect and flag any suspicious activities.

IV. METHODOLOGY AND ALGORITHM

Changes in breast surface temperature that could indicate abnormal tissue growth are captured by infrared thermography, commonly referred to as thermal imaging. AI-driven thermal image analysis uses CNNs (CNNs) and machine learning models to identify malignancies by analysing thermal trends. According to research with encouraging accuracy levels, thermal imaging is a viable nonsurgical diagnostic method. AI can evaluate thermal imaging, discern between benign and malignant tumours, and extract relevant data. Additionally, ML techniques can lessen the influence of environmental influences and standardize image capture.

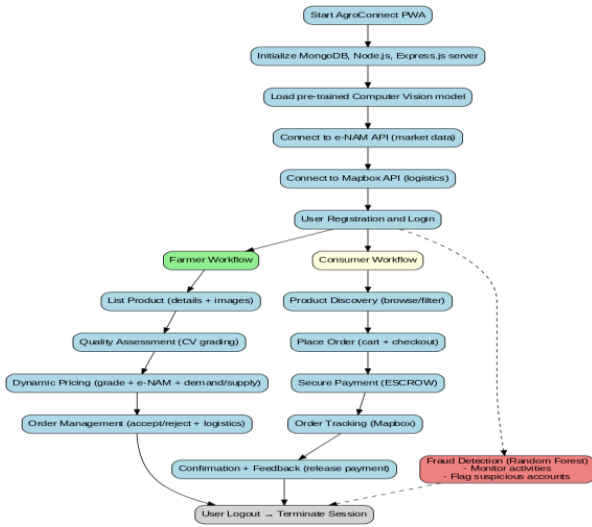


Fig. 2. System Flow Chart at Broader Level.

By lowering subjectivity in manual evaluations, CNNs, transformers, and generative adversarial networks (GANs) have greatly increased diagnostic accuracy. Additionally, they can spot minute trends that human observers might overlook, which could result in more precise diagnoses and prognoses.

A. Quality Verification Algorithm

The quality assessment system implements a multi-stage computer vision pipeline incorporating image preprocessing, feature extraction and classification components.

Image preprocessing applies standardization techniques including resizing to 224×224 pixels, normalization, and data augmentation to improve model robustness. U-Net segmentation isolates crop regions from background elements, reducing noise and improving classification accuracy.

The MobileNet CNN architecture balances computational efficiency with classification performance, enabling real-time processing on mobile devices. Transfer learning from ImageNet pretrained models accelerates training while maintaining high accuracy across diverse crop types.

Dynamic Pricing Engine

The pricing algorithm integrates multiple data sources to generate fair, market-responsive price recommendations.

$$P(t) = \alpha \cdot P_{\text{market}}(t) + \beta \cdot Q_{\text{score}} + \gamma \cdot D_{\text{local}}(t) + \delta \cdot S_{\text{seasonal}}(t)$$

Where $P(t)$ represents the recommended price at time t , $P_{\text{market}}(t)$ denotes current e-NAM market prices, Q_{score} represents the AI-generated quality score, $D_{\text{local}}(t)$ captures local demand patterns, and $S_{\text{seasonal}}(t)$ accounts for seasonal variations. Coefficients α , β , γ , and δ are optimized using historical transaction data and farmer feedback.

AgroConnect Platform Workflow

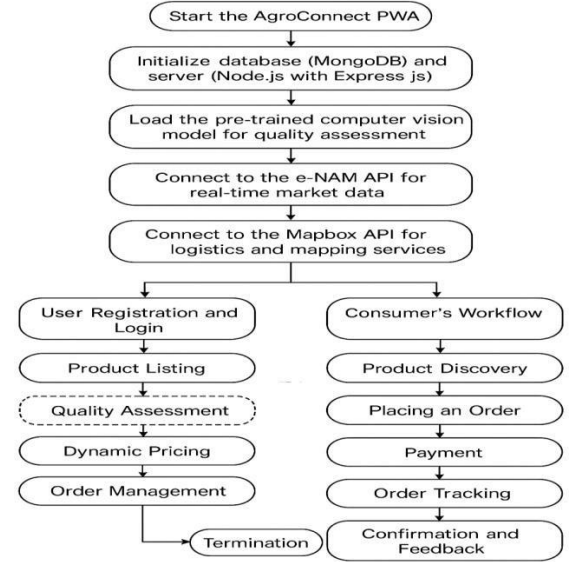


Fig. 3. Workflow at Implementation Level

B. Dynamic Pricing Engine

The pricing algorithm integrates multiple data sources to generate fair, market-responsive price recommendations.

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C. Dynamic Fraud Detection System

Random Forest algorithms analyze transaction patterns, user behavior and device characteristics to identify potentially fraudulent activities. Features include transaction velocity, geographic anomalies, device finger printing, and account age patterns.

SHAP (Shapley Additive exPlanations) interpretability ensures transparent decision-making, enabling manual review of flagged transactions and continuous model improvement based on false positive analysis.

V. IMPLEMENTATION

A. Core Features Implementation

- **Digital Storefronts:** Farmers create personalized profiles with Aadhaar verification, product listings, and inventory management capabilities. React components provide intuitive interfaces for product upload, pricing, and order management.
- **Real-time Price Integration:** Web Socket connections

maintain live price feeds from e-NAM APIs, updating product prices every 15minutes during market hours. Local caching ensures price availability during network interruptions.

- **Quality Grading:** Tensor Flow Lite models run client-side for immediate quality assessment, with server-side validation for dispute solution. Image compression and optimization reduce band width requirements while maintaining classification accuracy.

B. Performance Optimization

System performance optimization addresses the unique constraints of rural internet infrastructure and device limitations:

Network Optimization: Adaptive image loading, request batching and progressive enhancement ensure functionality across 2G/3G networks. Service Workers cache critical resources for offline operation.

Computational Efficiency: MobileNet architecture reduces inference time to <200ms on mid-range smartphones while maintaining >90%classification accuracy. Model quantization further reduces memory requirements

Technology Stack
leverages modern web technologies optimized for performance, scalability, and maintainability. The complete technology stack includes.

TABLE I. TECHNOLOGY STACK COMPONENTS

Layer	Technology	Purpose
Frontend	React.js, Redux, PWA	User Interface, State Management
Backend	Node.js, Express.js	API Services, Business Logic
Database	MongoDB, PostgreSQL	Data Storage, Transactions
AI/ML	Tensor Flow Lite, Scikit-learn	Quality Assessment, Pricing
Payments	Razorpay ESCROW,UPI	Secure Transactions
Infrastructure	AWSEC2, Firebase	Hosting, Mobile Services

VI. EXPERIMENTAL EVALUATION

A. Pilot Study Design

A comprehensive evaluation involved 500+ farmers and 200+ buyers across three agricultural regions in India: Punjab (wheat, rice), Karnataka (vegetables), and Maharashtra (cotton, sugarcane). The six-month pilot study compared AgroConnect transactions with traditional mandi systems across multiple metrics. Participants received training sessions on platform usage with ongoing support through multilingual help desk services. Control groups continued using traditional trading methods, enabling a comparative analysis of economic and operational outcomes.

B. Performance Metrics

Evaluation metrics encompassed technical performance, economic impact, and user experience dimensions.

TABLE II. SYSTEM PERFORMANCE METRICS

Metric	Target	Achieved	Improvement
Response Time	<3s	<2s	50%
Concurrent Users	5,000	10,000+	100%
Quality Accuracy	85%	92%	8.2%
Uptime	99%	99.7%	0.7%
Mobile Compatibility	90%	95%	5.6%



Fig. 4. Graphical Overview of System Performance.

VII. RESULTS AND DISCUSSION

A. Quantitative Outcomes

TABLE III. COMPARATIVE PERFORMANCE ANALYSIS

Metric	Traditional Mandi	AgroConnect	Improvement
Farmer's Price Share	25-30%	55-60%	~100%
Payment Disputes	15%	2.2%	85% reduction
Post-Harvest Loss	20-25%	8-10%	60% reduction
Transaction Time	3-5 days	<24 hours	>75% reduction

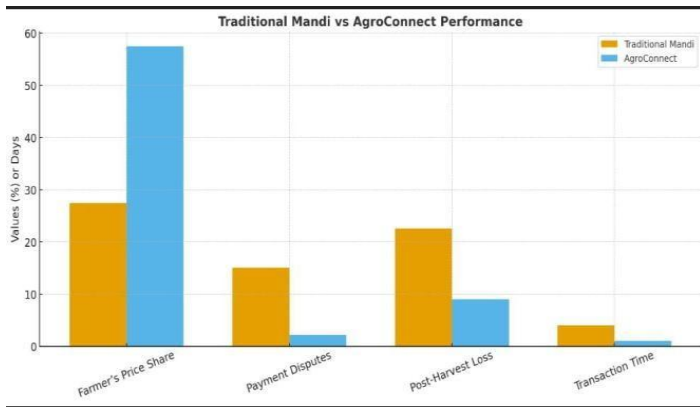


Fig. 5. Graphical Overview of Comparative Performance.

B. User Satisfaction Analysis

User satisfaction surveys revealed 92% overall satisfaction rates, with particularly high scores for transaction transparency (94%), payment security (91%), and quality verification reliability (89%). Farmers appreciated direct buyer communication (88% satisfaction) and real-time price updates (90% satisfaction). Buyer satisfaction centered on product quality consistency (87%) and delivery reliability (85%). The integrated logistics optimization reduced delivery times by 40% while maintaining cost competitiveness with traditional transportation methods.

C. Technical Performance

System scalability testing demonstrated stable performance under high-load conditions, supporting 10,000+ concurrent users with average response times below 2 seconds. The PWA architecture proved effective for rural connectivity, maintaining functionality during network interruptions and reducing data usage by 60%. AI model performance exceeded expectations, with quality classification accuracy reaching 92% across diverse crop types and lighting conditions. False positive rates remained below 5%, minimizing unnecessary manual reviews while maintaining quality assurance standards.

D. Challenges and Limitations

Implementation challenges included digital literacy gaps among older farmers, requiring extended training periods and ongoing support. Network connectivity limitations in remote areas occasionally impacted real-time features, addressed through offline functionality and data synchronization.

Quality assessment accuracy varied across crop types, with leafy vegetables showing lower classification performance (85%) compared to grains (95%). This limitation necessitates ongoing model training with expanded datasets for improved generalization.

VIII. CONCLUSION AND FUTURE SCOPE

AgroConnect demonstrates the transformative potential of integrated AI-driven platforms for agricultural commerce. The system successfully addresses key challenges in traditional supply chains while providing measurable economic benefits for farmers and buyers. Technical innovations in quality verification, dynamic pricing, and secure payments create comprehensive solution suitable for emerging market deployment.

Experimental validation confirms significant improvements in transaction efficiency, cost reduction, and user satisfaction. The platform's scalable architecture and technology choices position it for broader adoption across diverse agricultural contexts and geographic regions. Constraints in addition to concentrating on all future research

avenues.

Future development will incorporate advanced forecasting models using LSTM networks for demand prediction and inventory optimization. Drone-based quality verification systems will enable remote product assessment and expand service coverage to underserved regions.

Blockchain integration will provide immutable transaction records and supply chain traceability, addressing food safety concerns and export requirements. WhatsApp-based mini-applications will further reduce barriers to adoption for users with limited smartphone capabilities.

Integration with cold-chain logistics providers and expansion to processed food products represents significant commercial opportunities. Collaboration with government initiatives including PM-KISAN and FSSAI will accelerate platform adoption and ensure regulatory compliance.

AgroConnect contributes to UN Sustainable Development Goal2 (Zero Hunger) through improved market access, reduced food waste, and increased agricultural productivity. The platform's success demonstrates the potential for technology-driven solutions to address systemic challenges in agricultural development.

Open-source components and documented methodologies enable replication across different agricultural systems and emerging economies. The research contributes valuable datasets and algorithms to the agricultural technology community, supporting continued innovation in agriculture.

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